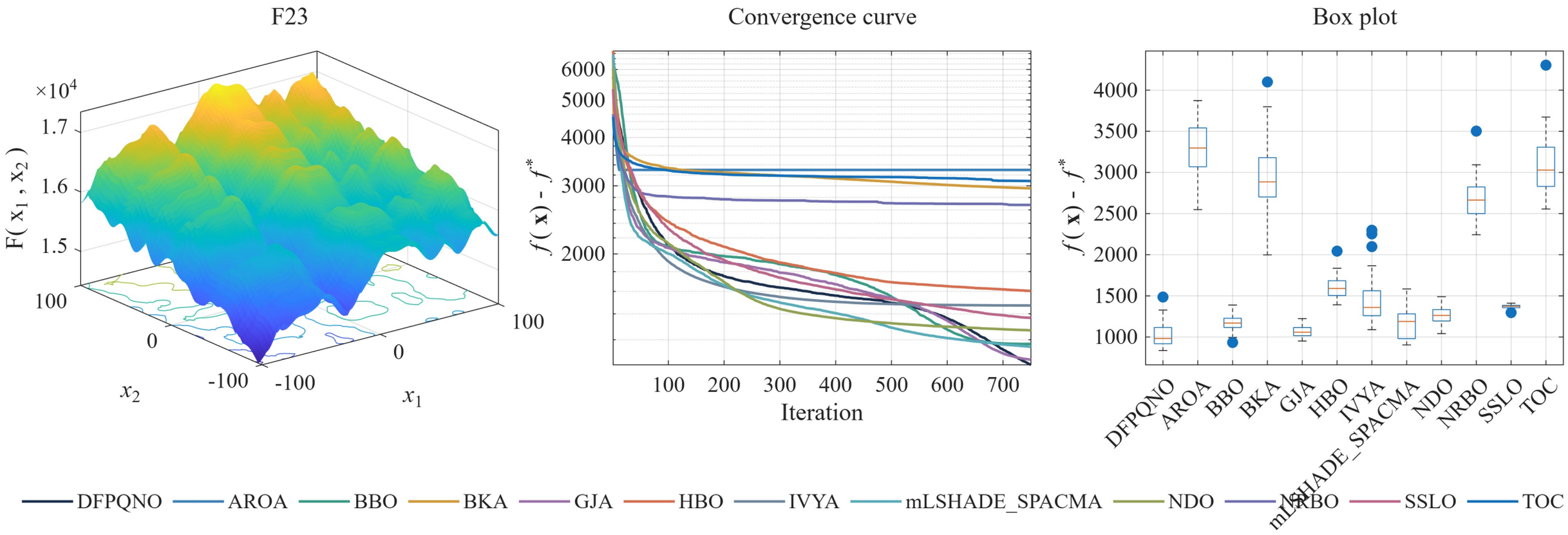

F23
×10^4
1.7
1.6
1.5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
6000
5000
4000
3000
2000
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
4000
3500
3000
2500
2000
1500
1000
f(x) - f*
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC